Uncertainty Estimates of Predictions via a General Bias-Variance Decomposition

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The Problem: Uncertainty Estimation Under Domain Drift

- **Context:** In safety-critical applications (e.g., cancer diagnostics), knowing *when* a model is wrong is as important as being correct.
- **Standard Practice:** Use the maximum softmax probability (confidence score). However, this is unreliable under **domain drift** (Ovadia et al., 2019).
- **Limitation of Existing Theory:** Proper scores (like Log-Likelihood) capture uncertainty, but lack a general bias-variance decomposition to quantify the *variance* of the prediction.

Research Question

Can we derive a general bias-variance decomposition for strictly proper scoring rules where the variance term naturally captures predictive uncertainty?

Related Work & Identified Gaps

Method / Work	**Limitation / Gap**
James & Hastie (1997)	General decomposition, but **no closed-form solution**.
Heskes (1998) / Hansen & Heskes (2000)	Specific to KL-divergence & Exponential Families only.
Domingos (2000)	Unclear if decomposition holds for arbitrary losses.
This work (Gruber & Buettner)	✓ General closed-form for proper scores. ✓ Unifies previous decompositions. ✓ Works in **logit space** (no normalization required).

Background: Bregman Divergences

Definition (Bregman Divergence)

Given a differentiable, convex function $\phi : U \to \mathbb{R}$, the Bregman divergence is:

$$d_\phi(x, y) = \phi(y) - \phi(x) - \langle \nabla\phi(x), y - x \rangle$$

Interpretation: Difference between $\phi(y)$ and the tangent at $\phi(x)$.

- Non-negative: $d_\phi \geq 0$, equality iff $x = y$.
- Generates Squared Error: If $\phi(x) = x^2$, then $d_\phi(x, y) = (x - y)^2$.

Background: Strictly Proper Scoring Rules

Definition (Strictly Proper Score)

A scoring rule $S(P, y)$ is *strictly proper* if the expected score is maximized iff the predicted distribution P equals the true data distribution Q:

$$\mathbb{E}_{y \sim Q}[S(Q, y)] \geq \mathbb{E}_{y \sim Q}[S(P, y)] \quad \forall P$$

- **Examples:** Log-Likelihood ($S(P, y) = \log p(y)$), Brier Score, Continuous Ranked Probability Score (CRPS).
- **Relation to Bregman:** Strictly proper scores correspond to Bregman divergences (Ovcharov, 2018).

$$G(Q) - S(P) \cdot Q = d_{G,S}(P, Q)$$

where G is the negative entropy.

Main Result: General Bias-Variance Decomposition

Theorem (General BVD for Proper Scores)

For a strictly proper score S with negative entropy G, estimated prediction $\hat{f}$, and target $Y \sim Q$:

$$\mathbb{E}[-S(\hat{f})(Y)] = \underbrace{-G(Q)}_{\textit{Noise}} + \underbrace{B_{G^*}[S(\hat{f})]}_{\textit{Variance}} + \underbrace{d_{G^*,S^{-1}}(S(Q), \mathbb{E}[S(\hat{f})])}_{\textit{Bias}^2}$$

- B_{G^*} is the **Bregman Information** generated by the convex conjugate G^*.
- The variance term generalizes the classical variance $\mathbb{V}[X]$ (since $B_{x^2}[X] = \mathbb{V}[X]$).

Definition (Bregman Information (Banerjee et al., 2005))

For a convex function ϕ and random variable X:

$$B_\phi[X] = \mathbb{E}[d_\phi(\mathbb{E}[X], X)] = \mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X])$$

Interpretation: Measures the **spread** of X around its mean. Generalization of Variance.

The Variance Term: Bregman Information

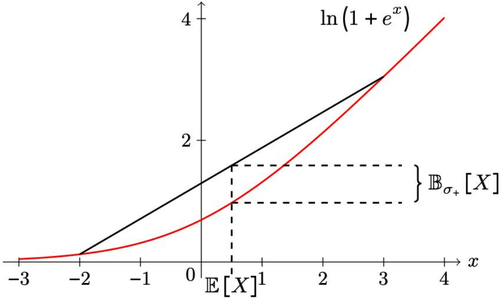


Figure 2: Illustration of the Bregman Information generated by the softplus function $\sigma_+(x) = \ln(1 + e^x)$ of a binary random variable.

Special Case: Exponential Families

Proposition (Exponential Family)

For an exponential family $p_\theta(y) = \exp(\langle \theta, T(y) \rangle - A(\theta)) h(y)$:

$$\mathbb{E}[-\ln p_{\hat{\theta}}(Y)] = n(\theta) + B_A[\hat{\theta}] + d_A(\theta, \mathbb{E}[\hat{\theta}])$$

where A is the log-partition function.

Implication:

- The variance term is $B_A[\hat{\theta}]$ (classical variance in the **natural parameter space** θ).
- Recovers the standard MSE decomposition (Example 3.5).

Special Case: Classification (Logit Space)

Corollary (Classification NLL Decomposition)

For logits $\hat{z} \in \mathbb{R}^k$ and softmax sm:

$$\mathbb{E}[-\ln sm_Y(\hat{z})] = H(Q) + B_{LSE}[\hat{z}] + d_{LSE}(sm^{-1}(Q), \mathbb{E}[\hat{z}])$$

where $LSE(x) = \ln \sum e^{x_i}$ is the LogSumExp function.

- **Surprising result:** The variance $B_{\mathrm{LSE}}[\hat{z}]$ measures uncertainty directly in the **logit space**.
- No normalization to probabilities required! This is highly practical for Deep Learning.

Application 1: Why Ensembles Reduce Uncertainty

- **Law of Total Variance for Bregman Information:**

$$B_G[X] = \mathbb{E}[B_G[X \mid Y]] + B_G[\mathbb{E}[X \mid Y]]$$

- **Interpretation for Ensembles:** Ensemble prediction $\hat{\theta}_D^{(n)}$ marginalizes out source of randomness W (e.g., initialization).
- As $n \to \infty$, $B_A[\hat{\theta}_D^{(n)}] \to B_A[\mathbb{E}[\theta_{W,D} \mid D]]$ (the variance due to W vanishes).
- **Conclusion:** Ensembles strictly improve expected performance (Equation 5).
- Using Markov's inequality on the Bregman divergence:

$$P\left(d_G(\mathbb{E}[X], X) \geq \frac{1}{\alpha} B_G[X]\right) \leq \alpha$$

- **Confidence Region:**

$$\mathrm{CR}_{(1-\alpha)} = \left\{x \mid d_G(\mathbb{E}[X], x) \leq \frac{B_G[X]}{\alpha}\right\}$$

Application 2: Confidence Regions via Markov's Inequality

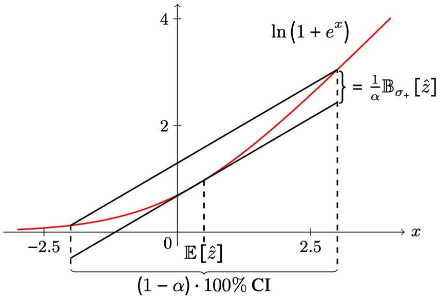


Figure 3: Illustration of the confidence interval for binary classification. The CI covers the area where the shifted tangent at $\mathbb{E}[\hat{z}]$ is larger than the generating convex function. This illustration generalizes to higher dimensions, where the CI is not an interval anymore in the strict sense but a region.

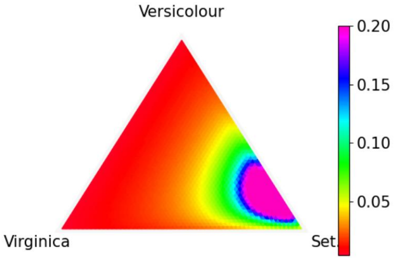
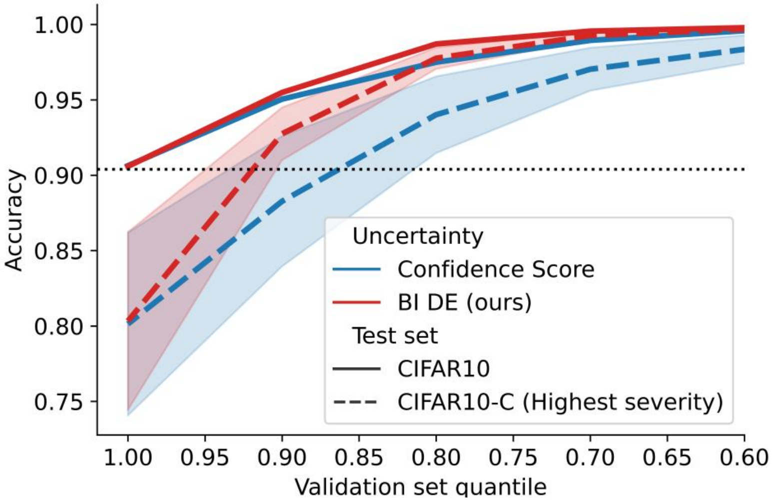


Figure 4: Confidence regions of a prediction transformed into the simplex for various alphas. The model is a simple neural network fitted on the Iris dataset.

Experiments: Out-of-Distribution Detection



Results: Common Classifiers on Toy Data

- **Setup:** Sampled multiple training sets to approximate ground truth Bregman Information (Section 4.1).
- **Observation:**
  - *High Bregman Info:* Decision Trees, k-NN (high variance due to data sensitivity).
  - *Low Bregman Info:* Random Forests, XGBoost (ensemble methods reduce variance).
  - Neural Networks exhibit moderate Bregman Info but are sensitive to initialization.
- **Qualitative insight:** The decomposition aligns with the intuitive notion of "model stability."

Discussion & Limitations

Strengths

- Unifies bias-variance theory under a single mathematical framework (Bregman).
- Provides practical tools (ensembles, confidence regions, OOD detection).
- Logit-space formulation is ideal for modern neural networks.

Limitations & Future Work

- **Computation:** Estimating Bregman Information requires multiple forward passes or an ensemble (computationally expensive).
- **Assumption:** Strictly proper scores only. Does not apply to 0-1 loss.
- **Future:** Extending to Bayesian Neural Networks and uncertainty quantification in LLMs.

Conclusion

- We introduced the first general **bias-variance decomposition for strictly proper scores**.
- The **Bregman Information** emerges as the universal variance term, generalizing classical variance.
- We derived practical formulations for **exponential families** and, crucially, **classification in the logit space** (LogSumExp).
- Demonstrated utility in:
  - Explaining the success of **ensembles**.
  - Building **confidence regions**.
  - Reliable **out-of-distribution detection** under domain drift.

Code: `https://github.com/MLO-lab/Uncertainty_Estimates_via_BVD`